

Synthesis and Reactions of Amines

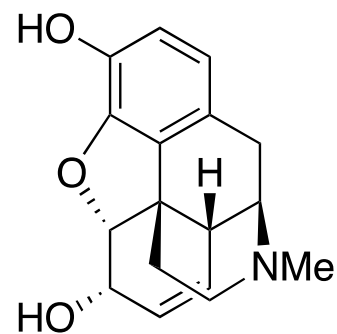
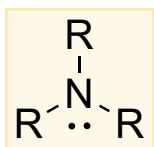
Organic Chemistry II

CHEM 2312

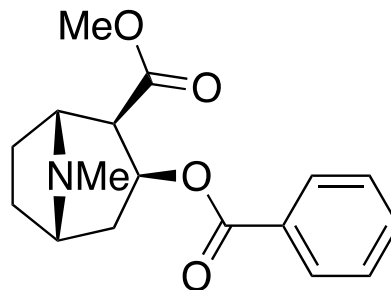
Dr. Michael Evans

1. Recognize the general structures of amines and classify them by substitution pattern.
2. Rationalize and apply the basicity of amines; rank a series of structurally related amines by basicity.
3. Predict products of reactions that yield amines and apply them in synthesis, including reductions of nitriles and alkyl azides, Gabriel reactions, and reductive aminations.
4. Describe reaction mechanisms involving amines and related compounds using elementary steps.
5. Predict products of reactions of amines with nitrous acid.
6. Apply aryl diazonium salts in synthesis and retrosynthesis as a useful tool for aromatic substitution reactions.

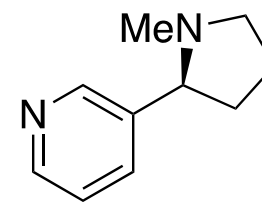
- Amines** are derivatives of ammonia in which one or more of the hydrogens has been replaced with alkyl or aryl groups



morphine

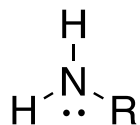


cocaine

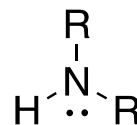


nicotine

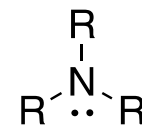
- Amine are classified as **primary (1°)**, **secondary (2°)**, or **tertiary (3°)** based on the number of carbon groups linked to the nitrogen atom



primary (1°)



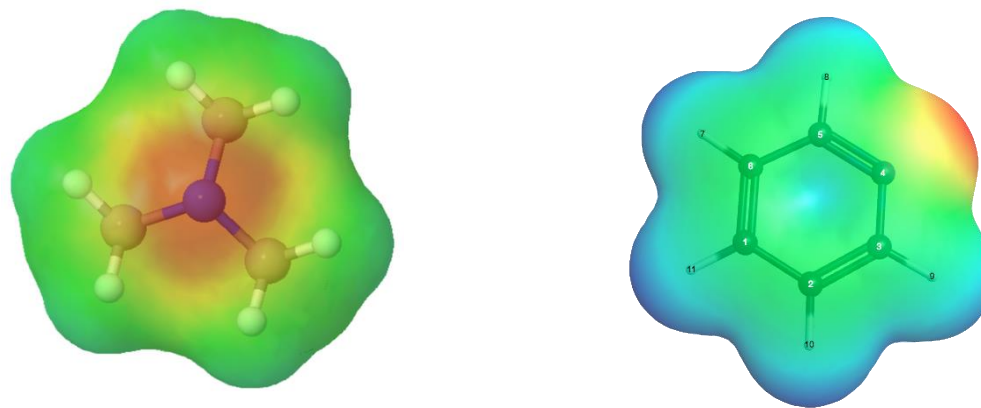
secondary (2°)



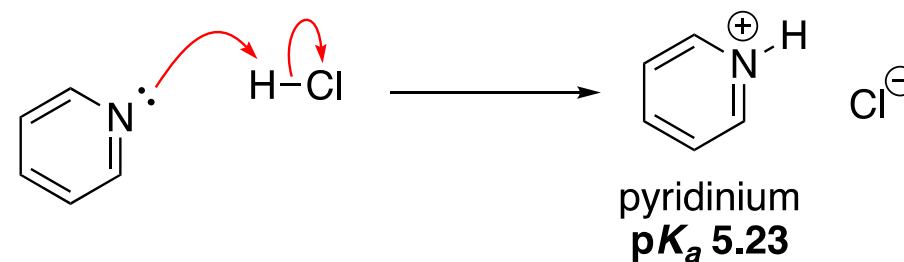
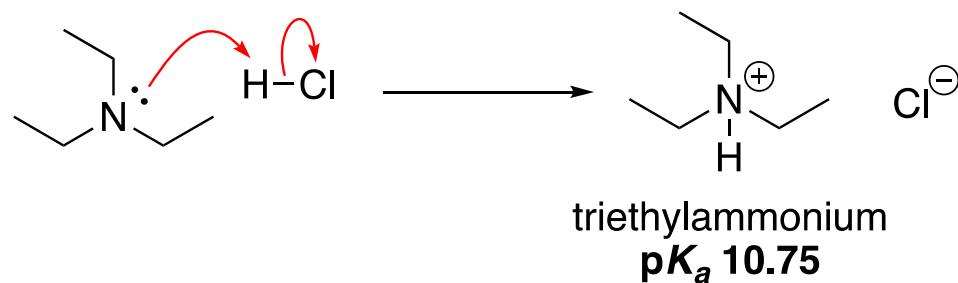
tertiary (3°)

22.1 Reactivity of Amines

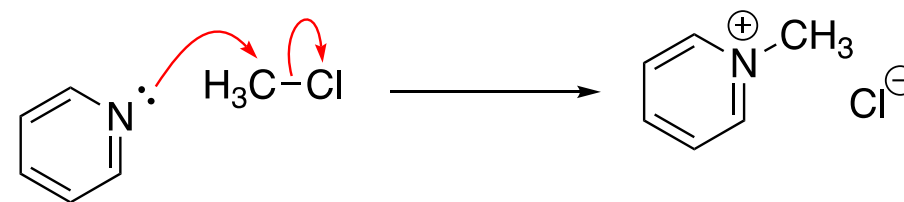
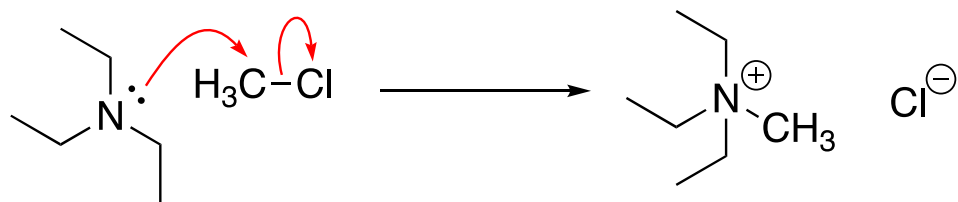
- The lone pair on nitrogen in amines is a very important Brønsted or Lewis base



- As a Brønsted base:

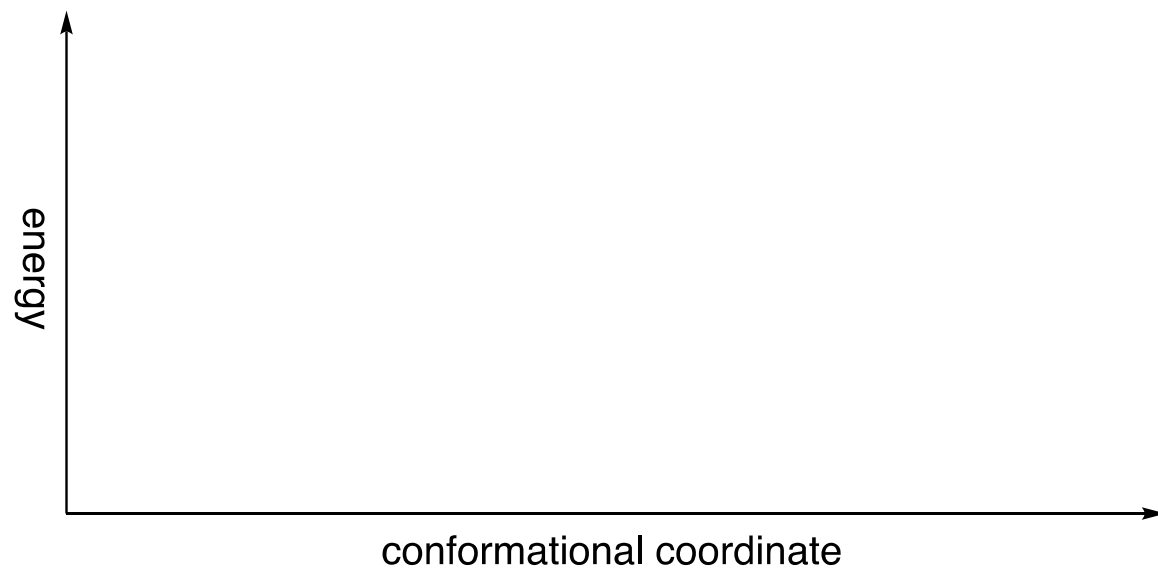
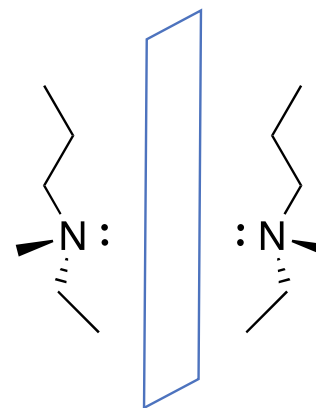
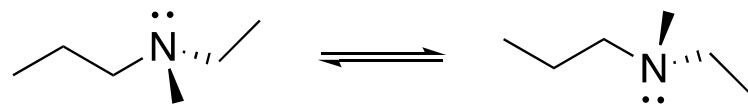


- As a Lewis base (nucleophile):



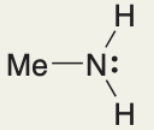
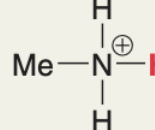
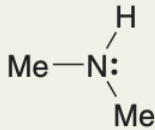
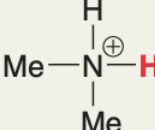
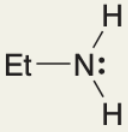
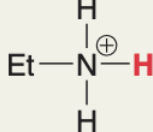
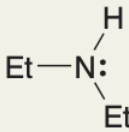
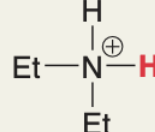
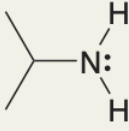
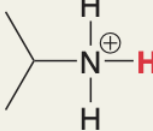
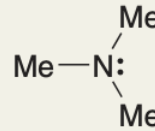
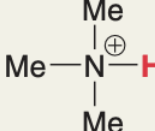
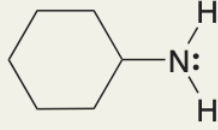
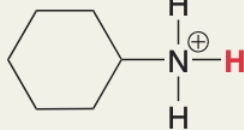
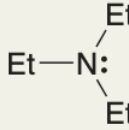
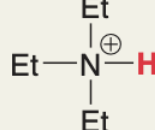
22.3 Chiral Amines Invert Rapidly

- Although amines with three different groups linked to nitrogen appear chiral, **amine inversion** via an sp^2 -at-nitrogen transition state is rapid at typical temperatures



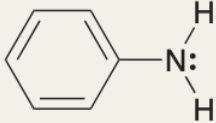
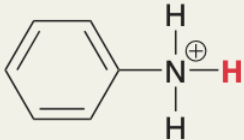
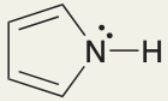
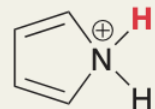
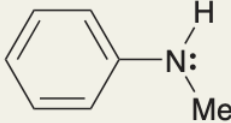
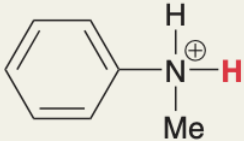
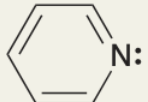
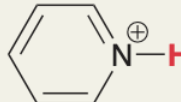
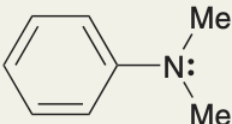
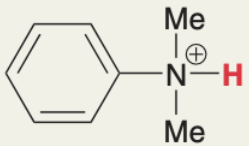
22.3 Brønsted Basicity of Amines | 1 of 2

- Alkyl ammoniums typically have acidity constants in the range of 9 – 11 (comparable to HCN and PhOH)

AMINE	pK _a OF AMMONIUM ION	AMINE	pK _a OF AMMONIUM ION
<i>Alkyl amines</i>		<i>Alkyl amines</i>	
 Methylamine	 pK_a = 10.6	 Dimethylamine	 pK_a = 10.7
 Ethylamine	 pK_a = 10.6	 Diethylamine	 pK_a = 11.0
 Isopropylamine	 pK_a = 10.6	 Trimethylamine	 pK_a = 9.8
 Cyclohexylamine	 pK_a = 10.7	 Triethylamine	 pK_a = 10.8

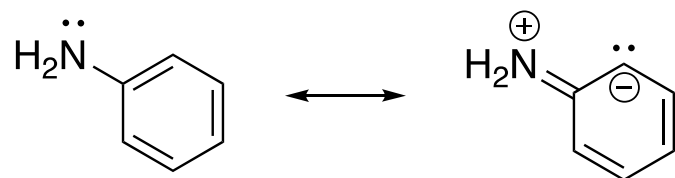
22.3 Brønsted Basicity of Amines | 2 of 2

- Aryl amines and aromatic heterocycles tend to be much less basic (five orders of magnitude or more!)

AMINE	pK_a OF AMMONIUM ION	AMINE	pK_a OF AMMONIUM ION
<i>Aryl amines</i>		<i>Heterocyclic amines</i>	
 Aniline	 $pK_a = 4.6$	 Pyrrole	 $pK_a = 0.4$
 N-Methylaniline	 $pK_a = 4.8$	 Pyridine	 $pK_a = 5.3$
 N,N-Dimethylaniline	 $pK_a = 5.1$		

22.3 Electron Delocalization in Anilines

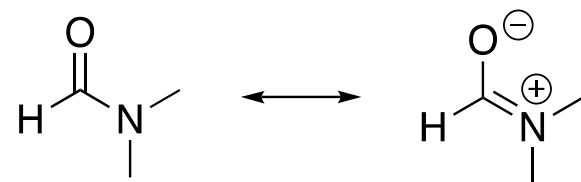
- The lone pair in anilines is highly delocalized
- Basicity is deeply affected by electron-donating groups and electron-withdrawing groups, especially in the *ortho* and *para* positions



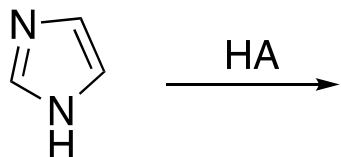
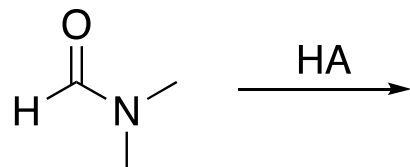
	<i>para</i> -Substituted aniline	Ammonium ion	
Increasing basicity ↑			Increasing acidity ↓

22.3 Electron Delocalization in Amides

- The lone pair on nitrogen in amides is similarly delocalized

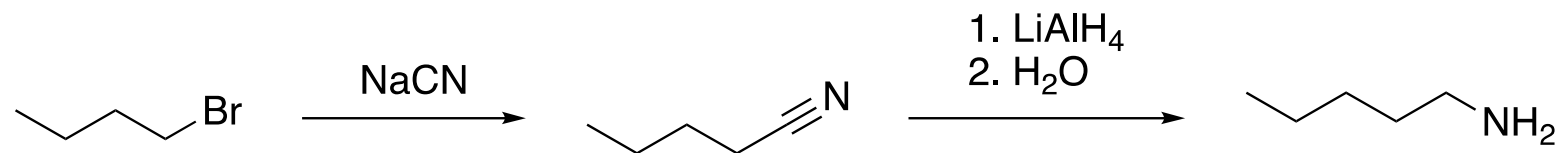


- The most basic atom in the amide group is *oxygen*, not nitrogen (compare to imidazole)

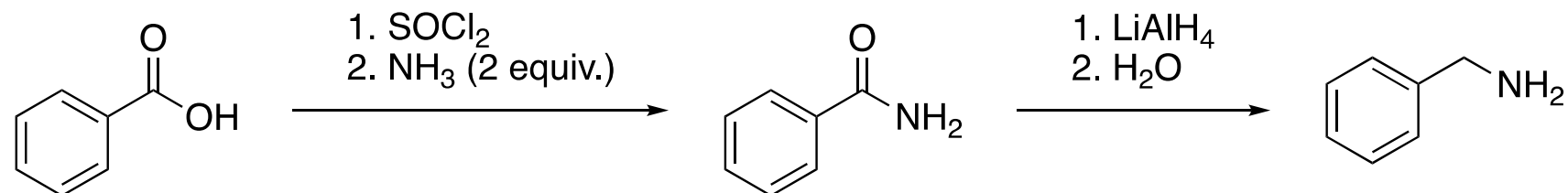


22.4 Previously Encountered Preparations of Amines

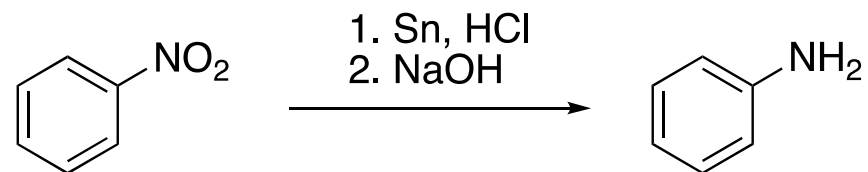
1. S_N2 substitution by cyanide (CN^-) followed by reduction with lithium aluminum hydride (LiAlH_4)



2. Reduction of amides by lithium aluminum hydride (LiAlH_4)

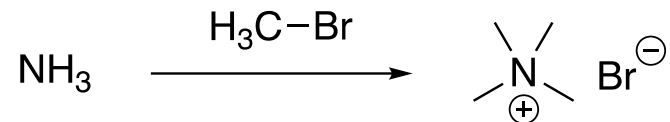


3. Reduction of nitrobenzenes to anilines



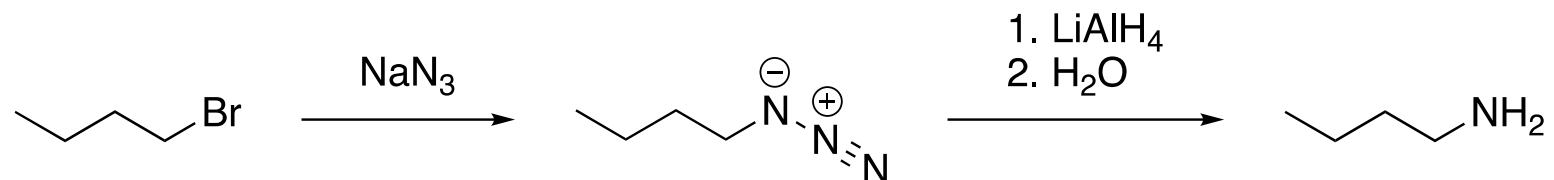
22.5 Alkylation of Ammonia Yields Quaternary Ammonium Salts *Only*

4. S_N2 alkylation of ammonia proceeds all the way to the quaternary ammonium salt in an uncontrollable manner. *DO NOT USE for synthesis.*

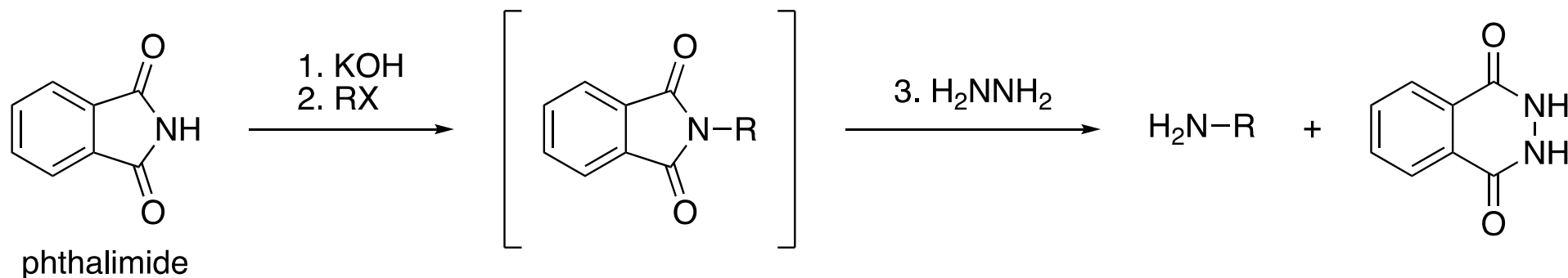


22.5 Preparations of Amines Using Ammonia Surrogates

5. S_N2 substitution by azide (N_3^-) followed by reduction with lithium aluminum hydride ($LiAlH_4$)

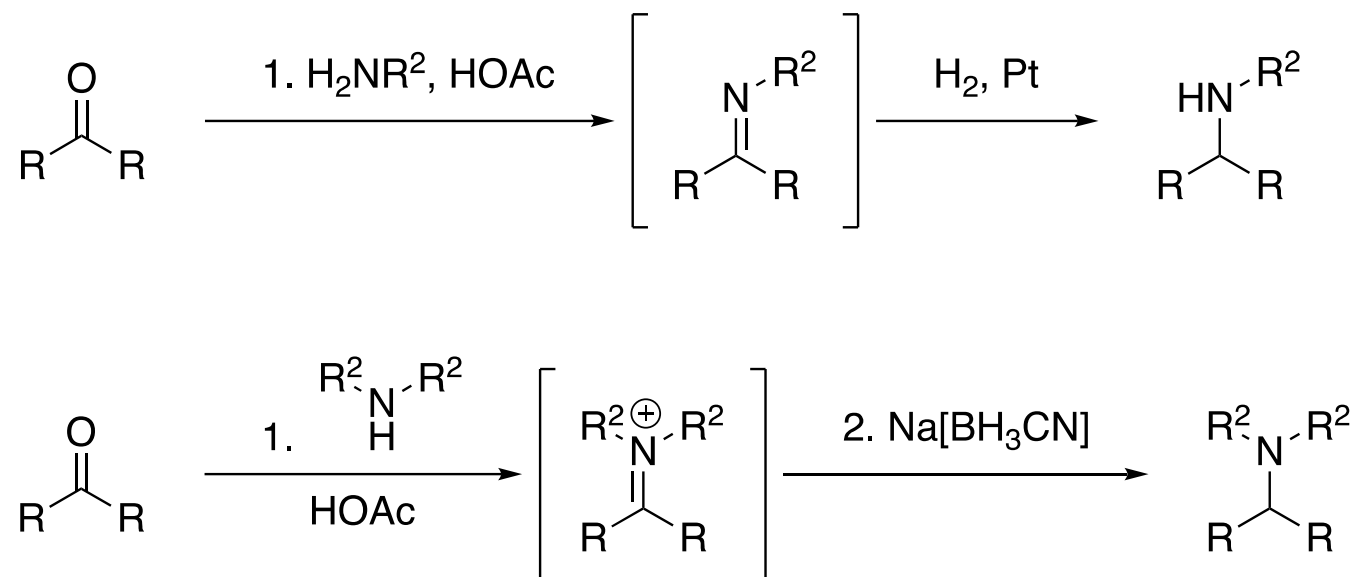


6. The **Gabriel synthesis**: alkylation of phthalimide anion followed by hydrolysis or hydrazinolysis

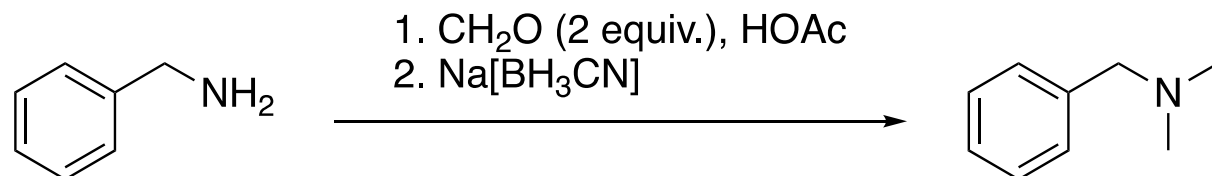


22.6 Preparations of Amines via Reductive Amination

7. **Reductive amination:** Condensation of an amine with a ketone or aldehyde followed by reduction with dihydrogen or (more common) sodium cyanoborohydride, $\text{Na}[\text{BH}_3\text{CN}]$



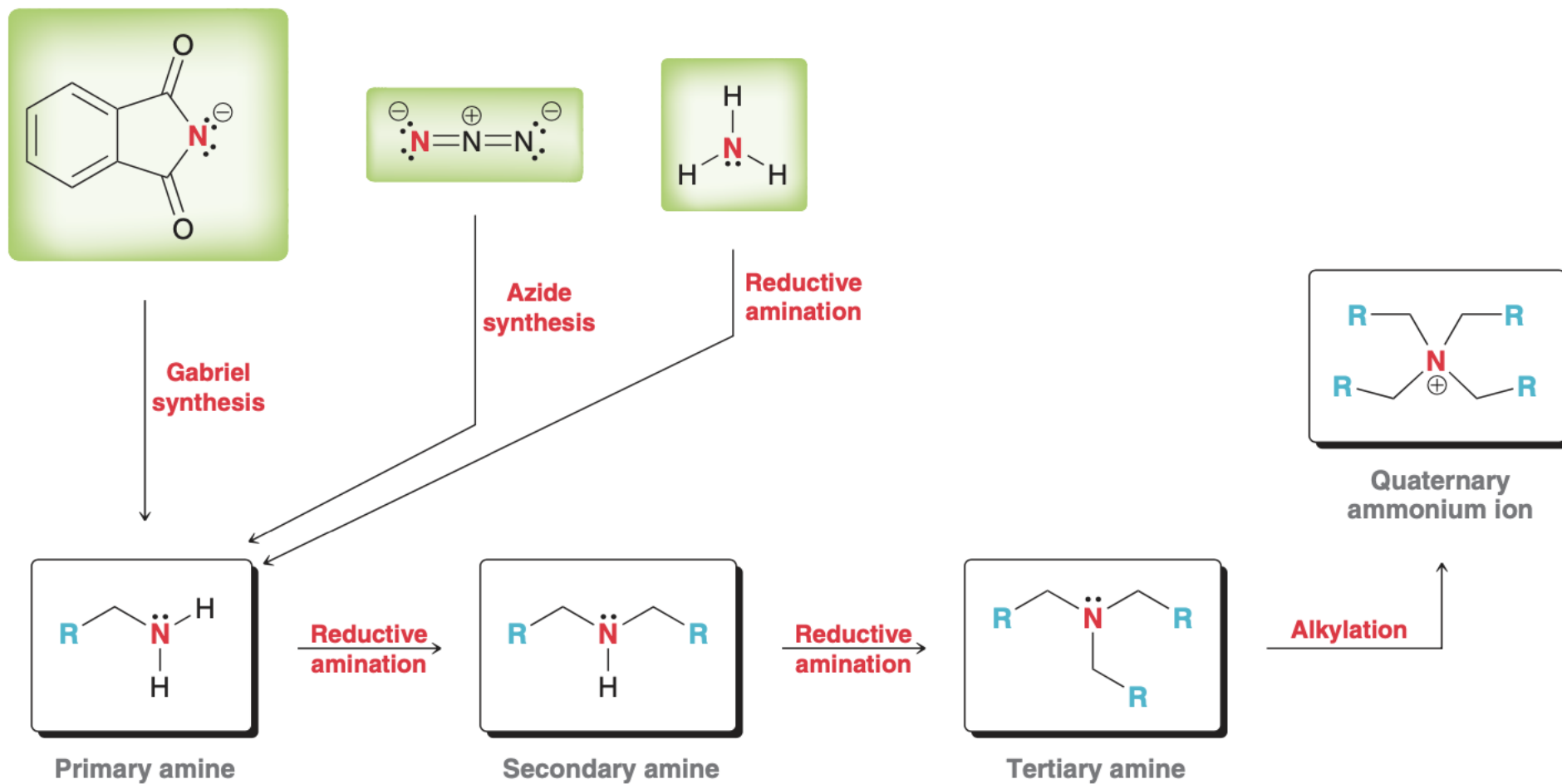
- Primary amines can be alkylated twice with two equivalents of ketohyde



8. Amides can be converted to amines via treatment with bromine (Br_2) and base (**Hofmann rearrangement**) followed by hydrolysis of the intermediate **isocyanate**

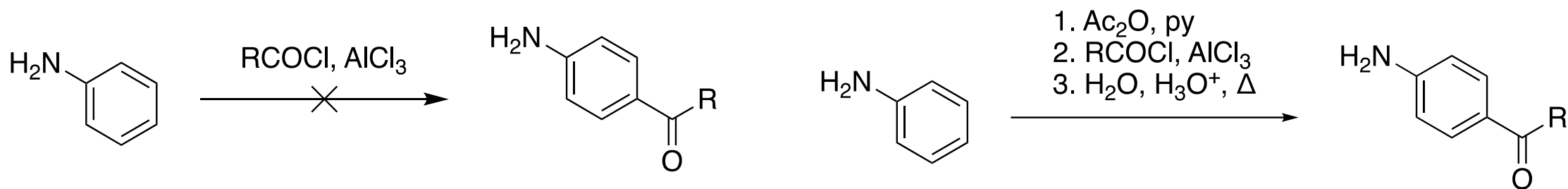
9. Acyl azides react similarly on heating (**Curtius rearrangement**), forming isocyanates that can be hydrolyzed to amides

22.7 Strategies for Synthesis of Amines

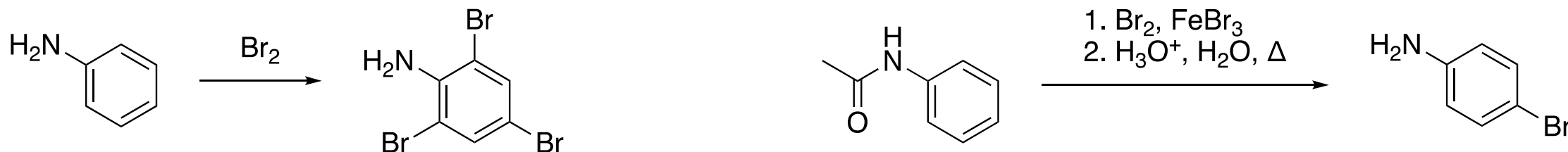


22.8 Acylation of Amines Tames the Basicity of the Nitrogen Lone Pair

- Acylation of an amino nitrogen converts the amino group into a much weaker base and weaker electron-donating group, enabling otherwise impossible reactions

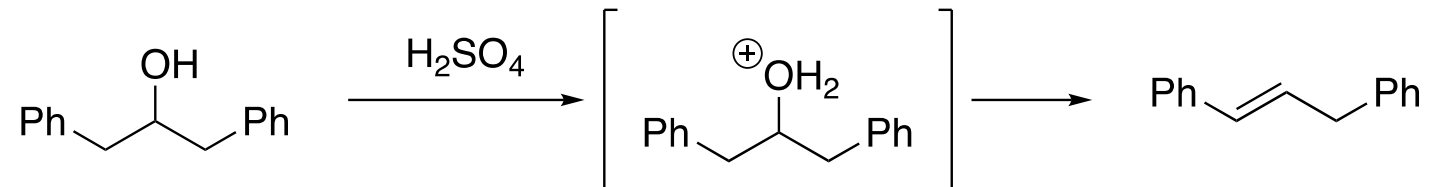


- *N*-Phenyl amides are much less reactive in electrophilic aromatic substitution than anilines

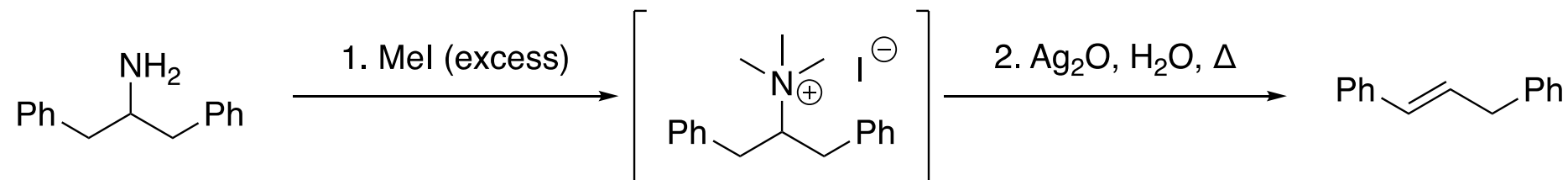


22.9 Hofmann Elimination: Amines as Leaving Groups

- Recall that alcohols undergo elimination to alkenes in the presence of strong acid (dehydration)

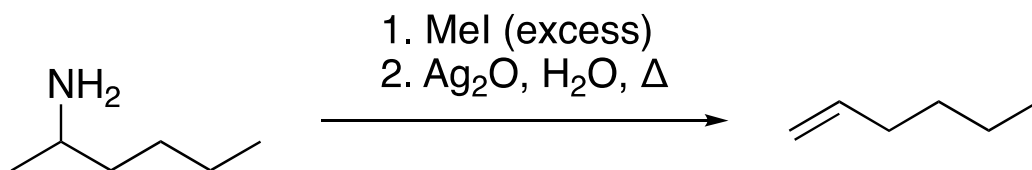


- Primary amines undergo a similar reaction in the presence of excess methyl iodide (CH_3I) with aqueous silver oxide as a base (**Hofmann elimination**)

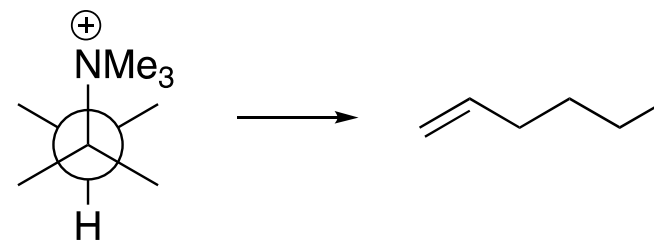
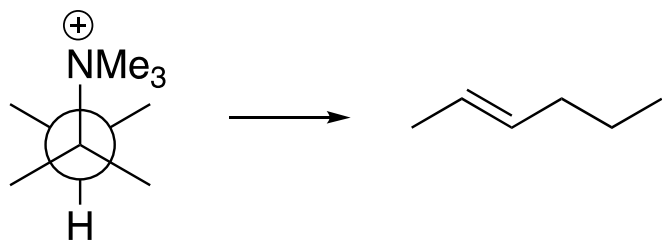


22.9 Hofmann Elimination Yields the Less Substituted Alkene

- When two alkenes with different substitution patterns are possible, the *less substituted* alkene is the major product

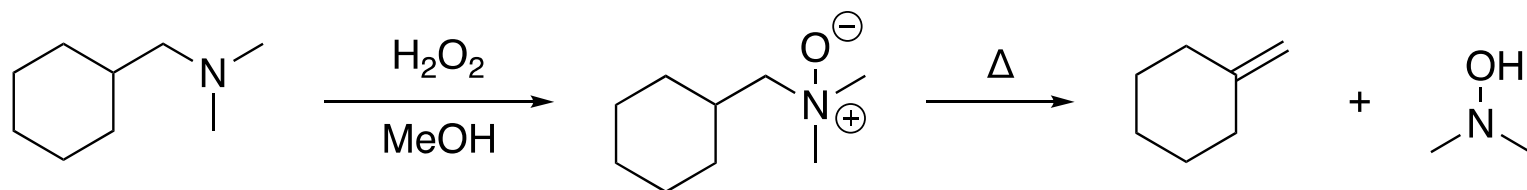


- Elimination at the less substituted β carbon avoids a destabilizing gauche interaction in the transition state for E2 elimination



Cope Elimination is a Concerted Reaction of *N*-Oxides

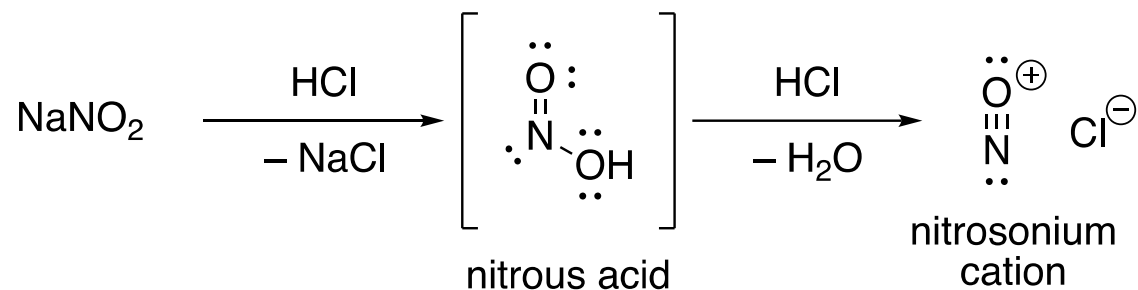
- Oxidation of amines with *meta*-chloroperbenzoic acid (*m*CPBA) yields ***N*-oxides**
- Heating *N*-oxides with appropriately situated β hydrogens results in concerted *syn* elimination of hydroxylamine (**Cope elimination**)



- The elimination involves intramolecular proton transfer and occurs in a single elementary step from the *N*-oxide

22.10 Nitrous Acid Forms the Nitrosonium Ion in Acidic Solution

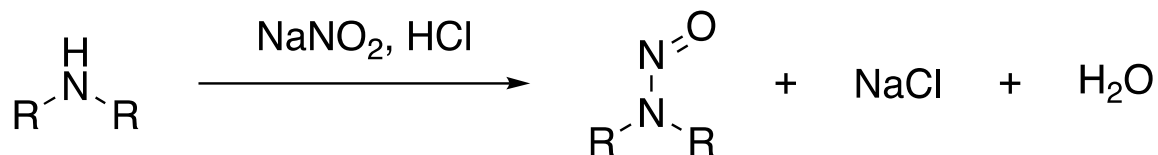
- Amines react with nitrous acid (HNO_2) to afford *N*-nitrosamines or diazonium salts
- Nitrous acid is generated *in situ* from an alkali metal nitrite salt and strong acid (HCl or H_2SO_4)
- Protonation and elimination of water gives the **nitrosonium cation**, a highly electrophilic species



- Compare to the mechanism of electrophilic aromatic nitration by HNO_3 , H_2SO_4

22.10 The Nitrosonium Ion Adds to Secondary Amines

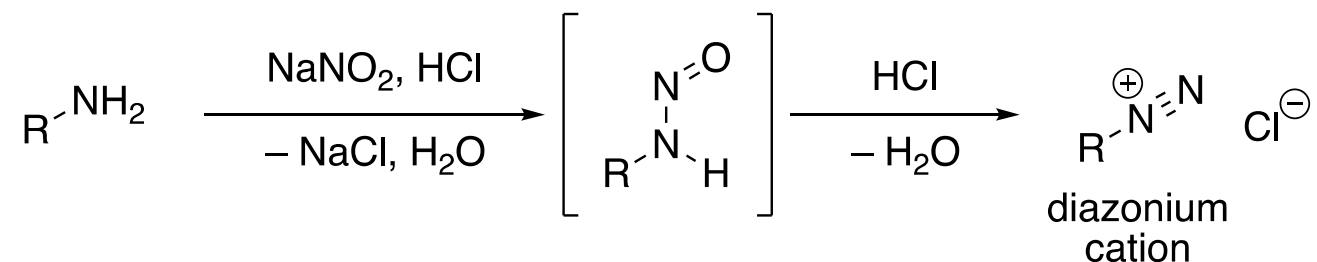
- Amines react with nitrous acid (HNO_2) to afford *N*-nitrosamines or diazonium salts
- Secondary amines add to the nitrosonium cation (1. Ad_N) to give ***N*-nitrosamines** after proton transfer (2. pt)



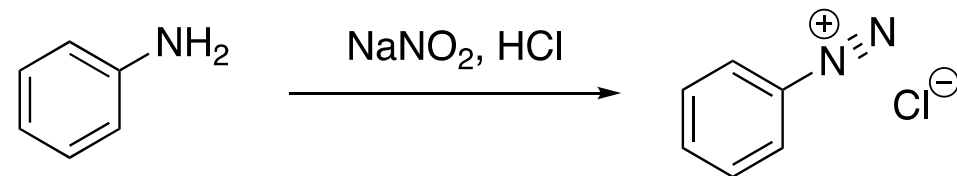
- N*-Nitrosamines are known to be potent carcinogens and can be generated in foods containing nitrites

22.10 The Nitrosonium Ion Condenses with Primary Amines to Form Diazoniums

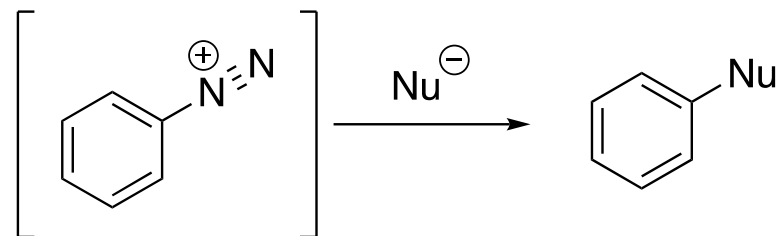
- Amines react with nitrous acid (HNO_2) to afford *N*-nitrosamines or diazonium salts
- Primary amines first form an unstable *N*-nitrosamine (1. Ad_N ; 2. pt) and then eliminate water (3. pt; 4. pt; 5. pt; 6. E_β) to give **diazonium salts ($\text{RN}_2^+ \text{X}^-$)**, a reaction known as **diazotization**



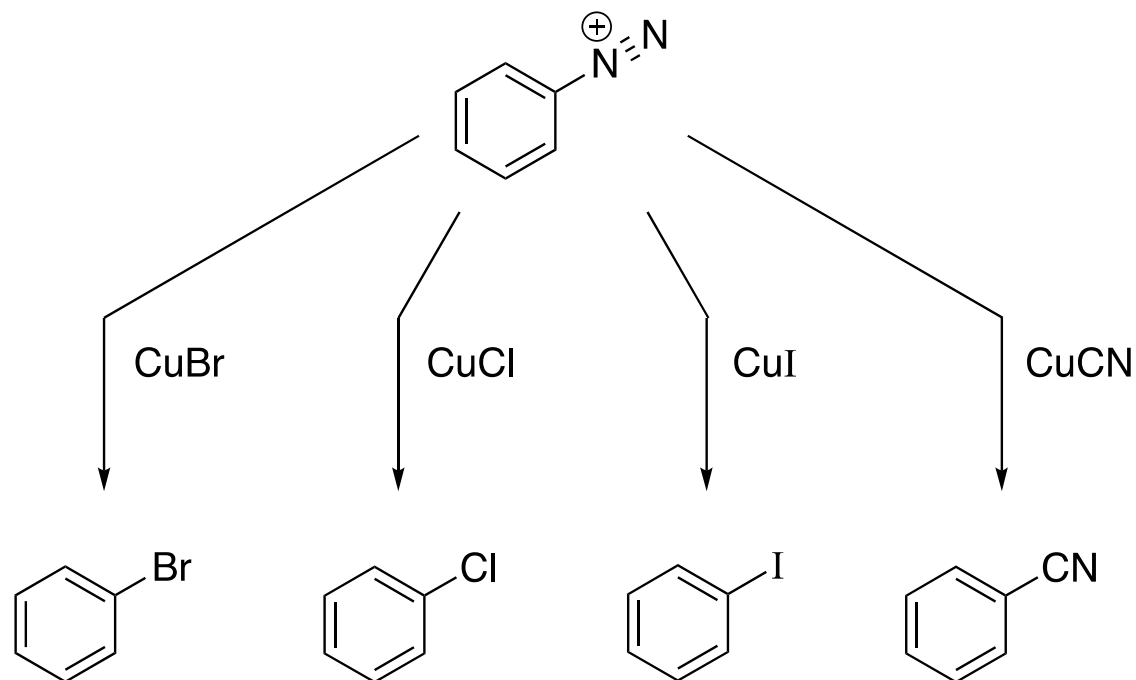
- Alkyl diazoniums are unstable and rapidly liberate N_2 , forming carbocations, but aryl diazoniums are more stable



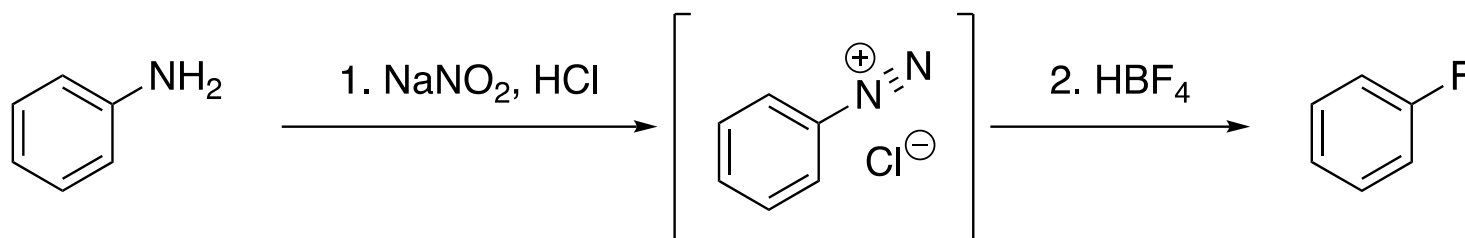
- Because N_2 is a fantastic leaving group, aryl diazoniums can participate in nucleophilic aromatic substitutions



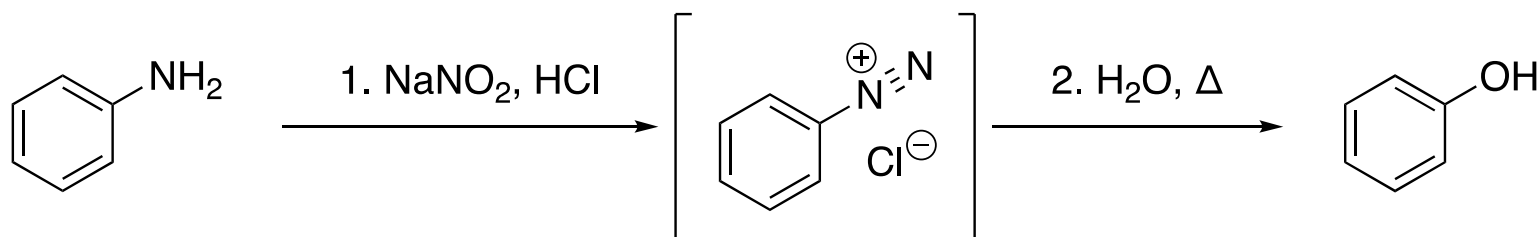
- Sandmeyer reactions** use copper(I) salts as the source of the nucleophile and install halogens or a cyano group



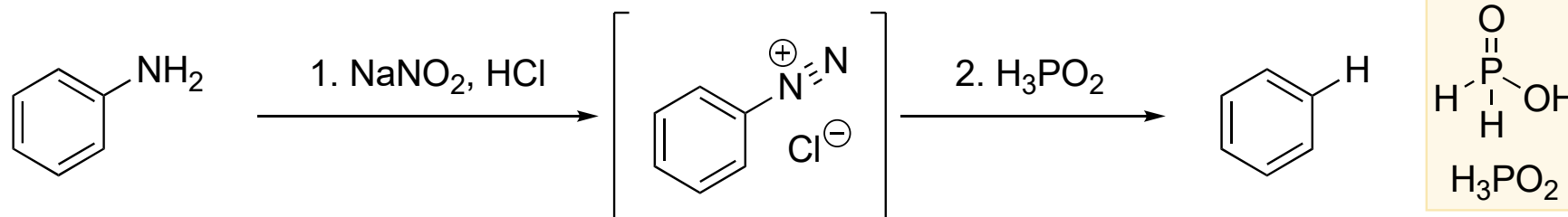
- Fluoroboric acid (HBF_4) serves as a source of fluoride anion in the **Balz-Schiemann reaction**



- Water displaces nitrogen gas when aqueous solutions of diazonium salts are gently heated

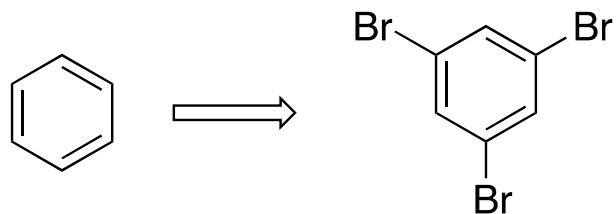


- Hypophosphorous acid (H_3PO_2) reduces the diazonium group to $-\text{H}$



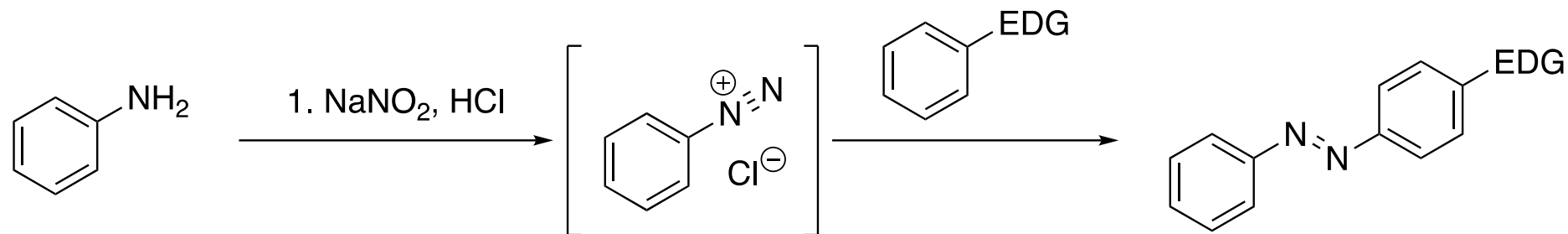
22.11 Applying Aryl Diazonium Salts in Synthesis

- How could we synthesize 1,3,5-tribromobenzene from benzene?



22.11 Aryl Diazoniums can Serve as Electrophiles in EAS Reactions

- In an **azo coupling** reaction, an aryl diazonium ArN_2^+ serves as the electrophile in an electrophilic aromatic substitution reaction with an electron-rich benzene



- How could we synthesize the **azo dye** below?

